THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The radius of a circle is 7 cm. The area of the circle is: 1

(a) 154 cm² **(b)** 22 cm² **(c)** 44 cm² **(d)** 308 cm²

2. The radius of a circle is 7 cm. The circumference of the circle is: 1

(a) 22 cm **(b)** 44 cm **(c)** 154 cm **(d)** 88 cm

3. The circumference of a circle is 44 cm. The radius of the circle is: 1

(a) 3.5 cm **(b)** 14 cm **(c)** 22 cm **(d)** 7 cm

4. The area of a circle is 154 cm^2. The radius of the circle is: 1

(a) 24.5 cm **(b)** 7 cm **(c)** 49 cm **(d)** 14 cm

5. A sector of a circle of radius 21 cm has central angle $60°$. The area of the sector is: 1

(a) 22 cm² **(b)** 462 cm² **(c)** 1386 cm² **(d)** 231 cm²

6. A sector of a circle of radius 21 cm has central angle $60°$. The length of its arc is: 1

(a) 64 cm **(b)** 22 cm **(c)** 44 cm **(d)** 231 cm

7. A semicircular protractor-shaped plate has radius 7 cm. The perimeter of the plate is: 1

(a) 58 cm **(b)** 29 cm **(c)** 22 cm **(d)** 36 cm

8. A quadrant of a circle has radius 7 cm. The perimeter of the quadrant is: 1

(a) 11 cm **(b)** 18 cm **(c)** 36 cm **(d)** 25 cm

9. The radius of a circle is 14 cm. The area of the circle is: 1

(a) 616 cm² **(b)** 88 cm² **(c)** 44 cm² **(d)** 1232 cm²

10. The radius of a circle is 14 cm. The circumference of the circle is: 1

(a) 176 cm **(b)** 88 cm **(c)** 44 cm **(d)** 616 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The radius of a circle is 3.5 m. Find the area of the circle. 2

12. The radius of a circle is 3.5 m. Find the circumference of the circle. 2

13. A sector of a circle of radius 21 cm has central angle $45°$. Find the area of the sector. 2

14. The radius of a circle is 10.5 m. Find the area of the circle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sector of a circle of radius 21 cm has central angle $45°$. Find the length of its arc. 3

16. A sector of a circle of radius 21 cm has central angle $60°$. Find the perimeter of the sector. 3

17. A semicircle has radius 7 cm. Find the area of the semicircle. 3

OR

A sector of a circle of radius 21 cm has central angle $72°$. Find the length of its arc. 3

18. A sector of a circle of radius 21 cm has central angle $90°$. Find the perimeter of the sector. 3

OR

A semicircle has radius 14 cm. Find the area of the semicircle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A circular park of radius 14 m is surrounded by a path 7 m wide. Find the area of the path. 5

OR

The largest possible circle is cut from a square sheet of side 14 cm. Find the area of the sheet left over. 5

20. A circular park of radius 21 m is surrounded by a path 7 m wide. Find the area of the path. 5

OR

The largest possible circle is cut from a square sheet of side 28 cm. Find the area of the sheet left over. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND